

Force of gravity

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In this exercise we implemented a Julia package which calculates the gravitational force between two parallel-placed unit homogeneous cubes at a distance of 1 (between the closest sides). Where the force between two bodies $T_1, T_2 \subset \mathbb{R}^3$ is equal to

$$\mathbf{F} = \int_{T_1} \int_{T_2} \frac{\mathbf{r}_1 - \mathbf{r}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|_2^3} d\mathbf{r}_1 d\mathbf{r}_2.$$

To do that we used the Gauss-Legendre quadrature, where we implemented both the calculation of nodes and weights using the Golub-Welsch algorithm.

1 Gravitational Force Integral

We have to calculate the gravitational force between two parallel-placed unit homogeneous cubes at a distance of 1.

We define the cubes as:

$$T_1 = [0, 1] \times [0, 1] \times [0, 1], \quad T_2 = [2, 3] \times [0, 1] \times [0, 1].$$

The second cube is shifted by 2 units in the x direction, so that the gap between the cubes is exactly 1.

We then expand the gravitational force integral:

$$\mathbf{F} = \int_{T_1} \int_{T_2} \frac{\mathbf{r}_1 - \mathbf{r}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|^3} d\mathbf{r}_2 d\mathbf{r}_1$$

where

$$\mathbf{r}_1 = (x_1, y_1, z_1), \quad \mathbf{r}_2 = (x_2, y_2, z_2),$$

into Cartesian coordinates which gives us the following six dimensional integral:

$$\mathbf{F} = \int_0^1 \int_0^1 \int_0^1 \int_2^3 \int_0^1 \int_0^1 \frac{(x_1 - x_2, y_1 - y_2, z_1 - z_2)}{[(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2]^{3/2}} dx_2 dy_2 dz_2 dx_1 dy_1 dz_1$$

Since the cubes are parallel and differ only in the x direction the force will therefore be:

$$\mathbf{F} = \int_0^1 \int_0^1 \int_0^1 \int_2^3 \int_0^1 \int_0^1 \frac{x_1 - x_2}{[(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2]^{3/2}} dx_2 dy_2 dz_2 dx_1 dy_1 dz_1$$

2 Gauss-Legendre quadrature

To calculate the 6 dimensional integral we used the Gauss-Legendre quadrature, which approximates a definite integral by a weighted sum

$$\int_a^b f(x) dx \approx \sum_{i=1}^n w_i f(x_i)$$

where x_i are the quadrature nodes and w_i are the corresponding weights.

2.1 Computing the nodes and weights

To find the nodes and weights we used the Golub-Welsch algorithm, which finds them as eigenvalues of the following tridiagonal matrix (Jacobi matrix)

$$J = \begin{pmatrix} 0 & b_1 & 0 & \cdots & 0 \\ b_1 & 0 & b_2 & \cdots & 0 \\ 0 & b_2 & 0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & b_{n-1} \\ 0 & 0 & \cdots & b_{n-1} & 0 \end{pmatrix}$$

where

$$b_i = \frac{i}{\sqrt{4i^2 - 1}}.$$

The eigenvalues λ_i of this matrix give the quadrature nodes, and the weights are computed from the first component of the normalized eigenvectors v_{1i} :

$$x_i w_i = 2(v_{1i})^2.$$

```
function my_gausslegendre(n)
    diagonal = zeros(n)
    subdiagonal = [i / sqrt(4i^2 - 1) for i in 1:(n-1)]
    J = SymTridiagonal(diagonal, subdiagonal)
    x, V = eigen(J)
    w = 2.0 .* (V[1, :].^2)
    return x, w
end
```

3 Demonstration

```

using dn2, Plots
n = 10
force_n = dn2.force(10)
rounded_force_n = round(force_n, digits=10)
println("Calculated force between two unit cubes one unit apart rounded to 10 digits:
$rounded_force_n")

force_n1 = dn2.force(11)
error = abs(force_n1 - force_n)
println("\nCalculated error: $error")

println("Plotting error drop off as we increase the number of nodes...")
ns = 2:n
vals = Float64[]
for _n in ns
    push!(vals, dn2.force(_n))
end

plot(ns, abs.(vals .- force_n1), ylabel="Error", xlabel="Number of nodes",
yscale=:log10, marker=:circle, legend=false, yticks=10.0 .^ (0:-5:-15))

```

Calculated force between two unit cubes one unit apart rounded to 10 digits
: 0.2479229692

Calculated error: 1.0547118733938987e-15
Plotting error drop off as we increase the number of nodes...

